

DBA5103

SA2 Group 4

**Minimizing Patient Waiting Time in Emergency Departments:
An Optimization Approach**



NUS
National University
of Singapore

Akash Mahanta (A0297661N)

Kulkarni Vrunda Vasant (A0297256N)

Ren Zijia (A0295902X)

Wu Chunzi (A0297302E)

1. Introduction

The emergency department (ED) often encounters challenges in managing patient flow, particularly during peak periods when demand surpasses available resources. These delays are frequently attributed to limited staffing, overcrowding, and inefficiencies in patient flow management. As patient demand increases amidst constrained healthcare capacity, EDs must navigate the delicate balance between addressing critical cases and attending to non-urgent patients, further exacerbating delays and straining resources.

This study aims to mitigate these issues by developing an optimization model to minimize patient waiting times through strategic alignment of nursing manpower with patient demand, particularly during peak hours. Effective waiting time management is essential for enhancing patient care quality, optimizing resource utilization, and strengthening the hospital's reputation. By leveraging mathematical optimization, this research seeks to improve patient experiences, health outcomes, and operational efficiency, while achieving cost savings for healthcare facilities.

2. Background

2.1 Importance of Reducing Waiting Time

Minimizing waiting times in EDs is vital for enhancing patient care and operational efficiency. Prolonged waits can worsen health outcomes for time-sensitive cases, diminish patient satisfaction, and erode trust in the healthcare system. Studies, such as those in the *Journal of Nursing Scholarship* [1], have linked overcrowding and extended wait times to higher mortality rates and adverse outcomes, while research in *BMJ Open Quality* [2] demonstrates that reducing cycle times improves patient satisfaction and care efficiency. Furthermore, extended waits increase costs for providers due to complications and prolonged stays. Systemically, reducing wait times optimizes resource utilization, bolsters hospital reputation, and improves community healthcare outcomes.

2.2 Previous Efforts and Methods

Efforts to reduce ED waiting times have focused on strategies like triage systems, optimized scheduling, resource reallocation, lean process redesign, and artificial intelligence. Triage systems enhance efficiency by prioritizing urgent cases, while scheduling algorithms and

strategic staffing address resource-demand mismatches. Lean methodologies reduce inefficiencies, and AI tools improve scheduling and predict patient flow, further decreasing delays. Although these approaches have improved satisfaction and efficiency, challenges such as implementation costs, training, and the need for adaptable solutions underscore the importance of context-specific approaches to healthcare optimization.

2.3. Overview of the Emergency Department

The Emergency Department (ED) operates through a structured process flow involving several prioritized areas. Patients first register at reception before proceeding with treatment. Patients requiring immediate attention are treated in the **Resuscitation** area, which has the highest priority. Resuscitation can happen at any stage in the patient journey at the ED. Registered patients are then triaged by nurses in the **Triage** area, given second priority. After triage, patients consult with doctors in the **Consultation** area, ranked third in priority. Following consultation and any necessary treatment, patients move to the **Observation** area, where nurses monitor their condition, which has the lowest priority. The process concludes with patients being either discharged or admitted to a ward for further care.

Emergency departments often experience high patient inflows, resulting in bottlenecks and overcrowding, especially during peak hours (typically, new patients arrive in ED during daytime from 9am-4pm). These issues stem from a combination of limited resources, including doctors, nurses, and available beds, as well as inefficiencies in patient flow management. However, for this paper, we are only focusing on optimizing nursing manpower to reduce waiting time. Typical shifts for nursing manpower are morning (AM), afternoon (PM) and night (ND) shifts.

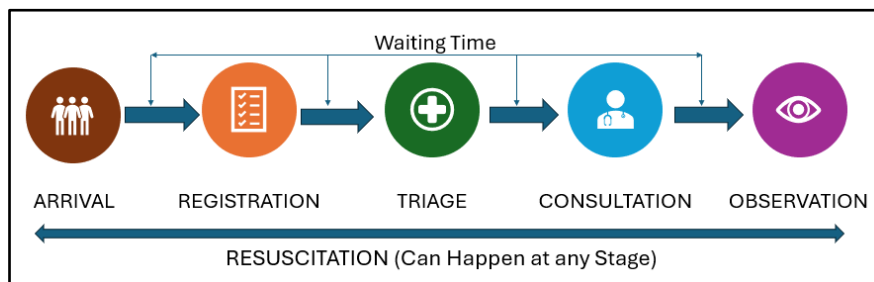


Figure 1: Patient journey diagram

3. Data Preparation

The MIMIC-IV-ED dataset serves as the primary input for this report. This dataset is a specialized subset of the MIMIC-IV (Medical Information Mart for Intensive Care) database, focusing specifically on emergency department (ED) visits.

According to official statistics from Singapore's Ministry of Health in 2024 [3], the average attendance of emergency department is around 400 (shown in Figure 2). Since this study only considers patients with arrival times before 6 PM, we proportionally select $400 * 18/24 = 300$ patients randomly from the dataset to represent a single day's data.

The selected features from the dataset include patient ID, arrival time (in time), and the emergency label. Additionally, each patient is assigned a numerical identifier based on their arrival time (in hours and minutes) to streamline optimization calculations. Figure 3 illustrates the distribution of patients.

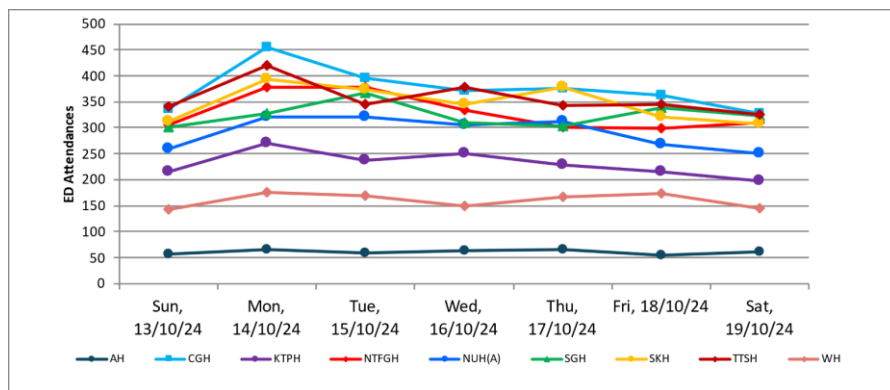


Figure 2: daily ED attendances (6 Oct 2024 – 12 Oct 2024)

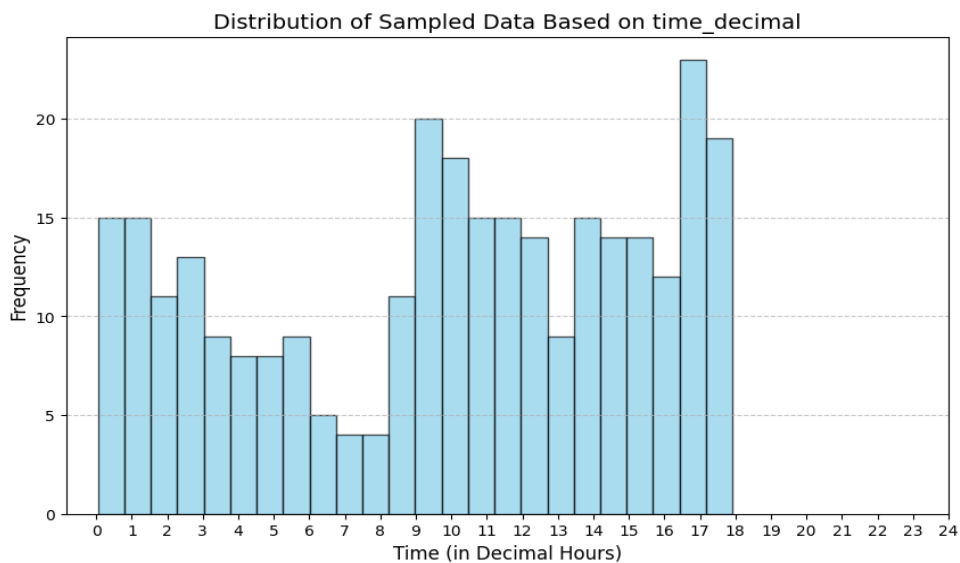


Figure 3: 300 Patients distribution

4. Methodology

4.1 Assumptions

1. We have $n=80$ nurses, and $m=300$ patients in one day
2. We only focus on how to assign nurses to different areas, in our model, we do not consider nurse shift rotations or the possibility of flexibly reallocating nurses from one area to another when they are idle.
3. 10% patients will need resus each day, and the resus happens before registration.
4. No patient passes away during this process
5. We define time in the form of hours. For example, 8:30 a.m. is represented as 8.5 hours. The time slots for each area are:

$$s_{resus} = 45 \text{ mins} = 0.75 \text{ hours};$$

$$s_{registration} = 5 \text{ mins} = \frac{1}{12} \text{ hours};$$

$$s_{triage} = 5 \text{ mins} = \frac{1}{12} \text{ hours};$$

$$s_{consultation} = 20 \text{ mins} = \frac{1}{3} \text{ hours};$$

$$s_{observation,end} = 120 \text{ mins} = 2 \text{ hours}$$

And we refer to the number of slots in each area as q .

$$q_{resus} = 32;$$

$$q_{registration} = 288;$$

$$q_{triage} = 288;$$

$$q_{consultation} = 72;$$

$$q_{observation,end} = 12$$

4.2 Objective Function, Decision Variables and Constraints

- **Objective function:** Minimize the waiting time of total patients

$$\min \sum_{i=1}^m \sum_{k=1}^5 waiting_time_{i,k}$$

- **Decision variables:**

1. $y_{j,k}$: A binary variable that equals 1 if nurse j is assigned to area k , and 0 otherwise.
2. $x_{i,a,k}$: A binary variable that equals 1 if patient i is in area k where k represents

resuscitation (1), registration (2), triage (3), consultation (4), or observation (5) during time slot a , and 0 otherwise.

3. $slot_start_time_{a,k}$: The start time of time slot a in area k .
4. r_i : A binary variable that equals 1 if patient i requires resuscitation, and 0 otherwise.
5. $arrival_{i,k}$: The time at which patient i arrives at area k .
6. $waiting_{time_{i,k}}$: The waiting time for patient i in area k before a nurse j becomes available.
7. $arrive_time_i$: The time at which patient i arrives at the emergency department (provided as input data).

Constraints:

1. Total Nurse Capacity: The total number of nurses in the ED at any time must not exceed n :

$$\sum_{j=1}^n \sum_{k=1}^5 y_{j,k} \leq n$$

2. Minimum Nurse Requirement per Area: Each area must have at least one nurse assigned at any time:

$$\sum_{j=1}^n y_{j,k} \geq 1, \text{ for } \forall k \in \{1,2,3,4,5\}$$

3. Single Area Assignment per Nurse: Each nurse can only be assigned to one area at a time:

$$\sum_{k=1}^5 y_{j,k} = 1, \quad \text{for } \forall j$$

4. Single Treatment per Patient per Area (Except Resuscitation): Each patient can only be treated once in each area, except for the Resuscitation area:

$$\sum_{a=1}^q x_{i,k,a} = 1, \quad \text{for } \forall i \text{ and } \forall k \in \{2,3,4,5\}$$

5. Capacity Constraint per Time Slot: The number of patients treated in a time slot cannot exceed the number of nurses available in that area:

$$\sum_{i=1}^m x_{i,k,a} \leq \sum_{j=1}^n y_{j,k}, \text{ for } \forall k \in \{1,2,3,4,5\} \text{ and } \forall a$$

6. Resuscitation Requirement: Whether a patient requires resuscitation is determined by r_i . A patient can only be treated in the Resuscitation area if $r_i = 1$.

$$x_{i,1,a} \leq r_i \text{ for } \forall i \text{ and } \forall a$$

7. Nurse and Patient assignment:

Assuming there are $n1$ nurses assigned to area k , and to simplify the problem, time has been divided into discrete slots. Each nurse is assumed to be available across the same set of time slots. In the absence of patients, all $n1$ nurses are available during each time slot. When a patient arrives, they are prioritized for assignment to the time slot closest to their arrival time, provided a nurse is available in that slot. However, even if nurses are available in the current time slot at the time of the patient's arrival, the patient will not be assigned to that slot to maintain alignment with the pre-defined scheduling rules.

For example: for resus part the slot table is shown

Time slot 1 (start time: 00:00): nurse 1, nurse 2, ...nurse $n1$
Time slot 2 (start time: 00:05): nurse 1, nurse 2, ...nurse $n1$
Time slot 3 (start time: 00:10): nurse 1, nurse 2, ...nurse $n1$
...
Time slot 288 (start time: 24:00)

If the patient comes at 00:05, we will assign him into time slot 2. If there are already $n1$ patients assigned into time slot 2, he will be assigned into time slot 3.

- (a) Define the arrive time of patient i in area k (big M here is used for patient who does not need resus, and $M = 1000000$)

$$arrival_{i,1} = arrivetime_i + (1 - r_i)M$$

$$arrival_{i,2} = s_1 \times r_i + \sum_{a=0}^q x_{i,1,a} \times slot_start_time_{a,1}, \text{ for } \forall i$$

$$arrival_{i,k} = s_{k-1} + \sum_{a=0}^q x_{i,k-1,a} \times slot_start_time_{a,k-1}, \text{ for } k \in \{3,4,5\}$$

(b) Patients should be assigned into the time slot later than their arrival time in area 2,3,4,5.

$$\sum_{a=0}^q x_{i,k,a} \times slot_start_time_{a,k} \geq arrival_{i,k}, \quad \text{for } \forall i \text{ and } \forall k \in \{2,3,4,5\}$$

(c) waiting time of each patient i is in area k is:

In area 1, if $r_i = 0$, then we have:

$$waiting_time_{i,1} = 0$$

In area 1, if $r_i = 1$, then we have :

$$waiting_time_{i,1} = \sum_{a=0}^q (x_{i,1,a} \times slot_start_time_{a,1}) - arrival_{i,1}, \text{ for } \forall i \in m$$

In area 2, 3, 4, 5 we have:

$$waiting_time_{i,k} = \sum_{a=0}^q (x_{i,k,a} \times slot_start_time_{a,k}) - arrival_{i,k}$$

$\text{for } \forall i \in m \text{ and } \forall k \in \{2,3,4,5\}$

8. Patients should be treated in order:

$$slot_start_time_{i,a,k} t_{a,k} \geq slot_start_time_{i+1,a,k} t_{a,k}$$

5. Analysis and Results

5.1 Model implementation

Due to the large number of decision variables, such as when the number of nurses is set to 75, the model still includes around 100000 variables even after Gurobi's automatic presolve process. To enhance the optimization speed, we set parameter 'MIPGap' to value 0.01, allowing the solver to accept the solutions within 1% of the optimal objective value.

5.2 Results

5.2.1 Relationship between nurses and waiting time

In our experiment, we aim to find the relationship between the number of nurses and patients' waiting time. By testing different number of nurses, we obtained the following results:

Table 1: Variation of minimizing waiting time with number of nurses

<i># of nurses</i>	Total waiting time (hours)	Average waiting time (hours)
<i><50</i>	Infeasible	Infeasible
<i>50</i>	440	1.47
<i>55</i>	390	1.30
<i>60</i>	374	1.25
<i>65</i>	360	1.20
<i>70</i>	352	1.17
<i>75</i>	350	1.17
<i>80</i>	348	1.16
<i>90</i>	348	1.16
<i>100</i>	348	1.16
<i>110</i>	348	1.16
<i>120</i>	348	1.16

From table 1, it is evident that the average waiting time decreases as the number of nurses increases. When the number of nurses falls below 50, no feasible solution can be achieved, this indicates that under our assumptions, if the number of nurses is insufficient, not all patients can be treated within the same day, and some of them will need to wait until the following day for treatment. Even the latest-arriving patient reaches the ED before 6 PM.

Figure 4 illustrates the trend between increasing number of nurses and decreasing average waiting time. From the graph, as the number of nurses uniformly increases, the average waiting time reduces sharply at first and slows down. When the number of nurses reaches 80, the waiting time stabilizes, and the average waiting time is 1.16 hours per patient.

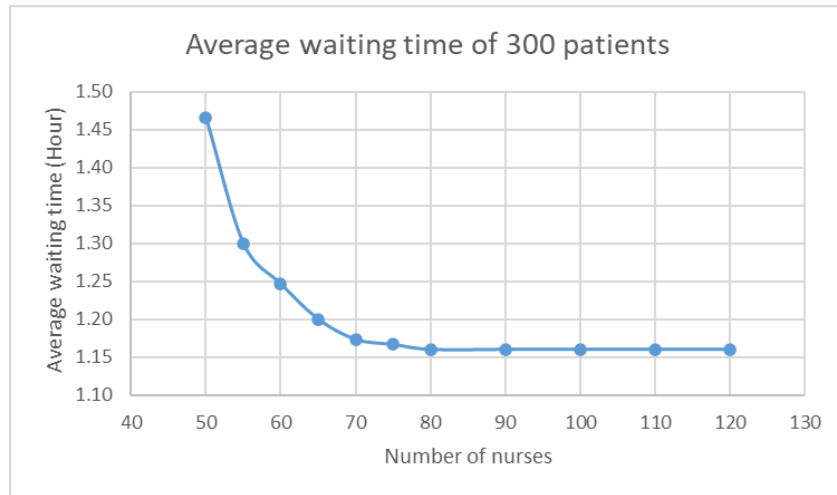


Figure4: Average waiting time of 300 patients

5.2.2 Nurses assignment

Figure 5 illustrates the relationship between the total number of nurses and the distribution of nurse allocations across various areas.

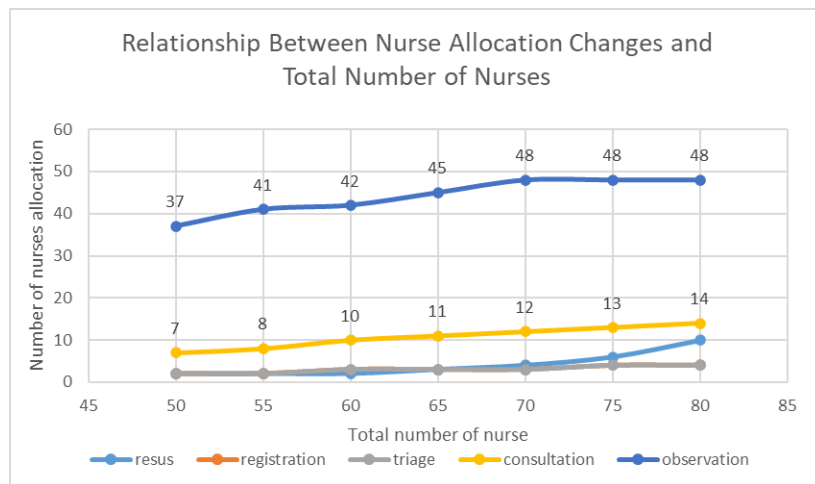


Figure 5: Relationship between nurse allocation changes and total number of nurses

Due to there is no change of waiting time when the number of nurses greater than 80, Figure 3 here only shows nurse allocation changes from 50 to 80. As the total number of nurses increases, number of nurses in observation stage increases from 37 to 48, reflecting the primary area of focus. In the other words, when the total number of nurses increases by 30, more than 30% of nurses are allocated to observation area. The number of nurses in consultation shows a gradual but smaller increase. This trend suggests that while the increase in nurses leads to a proportional rise in allocation across all areas, the majority of resources remain concentrated in observation.

5.2.3 Total waiting time in different areas

The waiting times for the resuscitation, registration, and triage phases are almost negligible. Most of the waiting time occurs in the observation phase, which has the longest treatment duration (Figure 6). However, even though we add more nurses in the observation area, the main reduction occurs in consultation area. The reason will be interpreted in the next part.

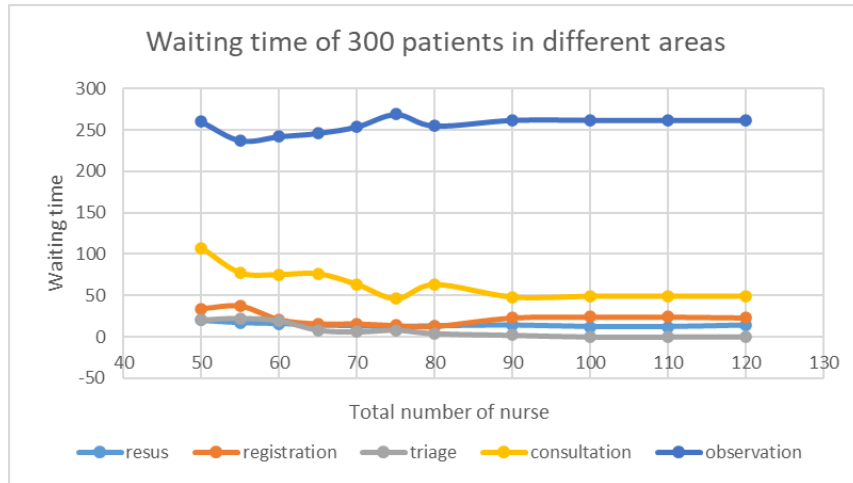


Figure 6: Waiting time of 300 patients in different areas

5.3 Interpretations

5.3.1 Unable to eliminate waiting time

Intuitively, when we have enough nurses, we should be able to reduce waiting time to zero. However, our experiment did not achieve this. That is because in our model, for the sake of simplified calculations, we divided the nurses' working hours into fixed time slots based on the treatment duration of each stage. Each time slot has a designated start time, and when a patient arrives at a particular area, they can only receive treatment in a time slot that starts after their arrival time. This means that even if there are nurses available in an ongoing time slot when the patient arrives, they cannot provide treatment for the patient. In other words, we "force" the patient to wait until the next time slot begins before receiving treatment. This approach inevitably results in some wasted nursing resources and unavoidable waiting times.

5.3.2 Non-linear relationship between nurses and waiting time:

The relationship between the number of nurses and the reduction in waiting time is not linear due to diminishing marginal returns. The possible reason is that, in the experiment, as shown in Figure 3, patients' arrivals are not evenly distributed throughout the day. Even if more nurses

are available, there might still be peak times when demand exceeds capacity, limiting the reduction in waiting time.

5.3.3 ‘Unchanged’ waiting time in observation

In the previous section, we were surprised to find that despite allocating more staff to the observation area, the waiting time in this area did not decrease. Two key reasons contribute to this outcome:

- Observation requires the longest treatment time among all steps. Even with a limited number of nurses (e.g., 50 nurses), a significant portion of the available staff is allocated to this area to handle its workload.
- Observation is the last step in the patient flow. With more nurses, the waiting times in earlier stages, such as consultation, are reduced. This means that more patients complete the earlier steps more quickly and then wait for treatment in the observation area. Therefore, we also need to increase the number of staff in the observation area to ensure that patients do not pile up in this step.

5.3.4 Resource prioritization for time-intensive stages

Our model arranges more nurses in areas with the longest treatment times, which is the observation stage. It is a strategic decision to minimize patient waiting time and improve overall efficiency. That is because we want to reduce bottlenecks. Areas with long treatment times are potential bottlenecks in the system.

6. Discussion

6.1 Limitations:

As we mentioned in the second part, the conditions are much more complex than what we do in this report. The main limitations in our model are:

- **Fixed time slot:** We divided nurses' working hours into fixed time slots, and patients can only be treated in the next available time slot upon arriving at a certain area. This approach leads to underutilization of nursing resources and results in unnecessary waiting times.
- **Fixed resuscitation part:** To simplify the calculations, we arranged the resuscitation

process at the very beginning. However, resuscitation in the emergency department can occur at any time.

- **No shift allocation:** Our model does not consider nurse shift schedules. There are typically 2 or 3 shifts per day. Additionally, hospitals must consider factors such as the maximum weekly working hours for each nurse, restrictions on consecutive night shifts, and other related issues.
- **No flexible allocation:** In our model, nurses are required to be assigned to a fixed area. However, in real-life scenarios, some hospitals allow for flexible nurse allocation, where nurses from different areas can provide mutual support.

6.2 Future Improvement:

- **Dynamic Time Slot Allocation:** Implement dynamic time slot allocation based on patient arrival times and current nurse availability. This could improve the utilization of nursing resources and reduce unnecessary waiting times.
- **Realistic Resuscitation Modeling:** Incorporate flexibility in the timing of the resuscitation process to reflect real-world scenarios where resuscitation may occur at any time. This adjustment would make the model more practical and applicable to actual emergency department workflows.
- **Incorporating nurse shift schedules:** Add shift allocation constraints to the model to better align with real-life nurse scheduling practices. Consider maximum weekly hours, night shift rules etc.
- **Flexible nurse allocation across areas:** Introduce flexibility in nurse allocation, allowing nurses to support multiple areas as needed. This would optimize resource utilization during peak times or emergencies.

6.3 Practical recommendations:

In the actual Singapore healthcare system, the typical nurse-to-bed ratio for general wards in the public acute hospitals is one nurse for every four or five beds. [4] Our optimization results for minimizing patient waiting time is 80 nurses for 300 patients arriving before 6 PM. The corresponding nurse-to-bed ratio in our case is about 1: 3.75, which indicates one nurse is

allocated for 3.75 patients.

Even we construct an idealized and simplifier scenario for calculation, there are still practical recommendations that can be applied to real-world situations.

- **Prioritize resource allocation:** Analyze the amount spent at each stage and allocate more resources, such as nurses, to stages or areas where the treatment time is the longest to reduce bottlenecks and overall waiting time.
- **Flexible nurse allocation:** Analyze the arrival patterns of emergency room patients, including daily peak times, weekends, and holidays, and flexibly allocate nurses based on these distribution patterns.
- **Diminishing returns analysis:** In reality, with a limited budget, it is crucial to evaluate the diminishing returns of adding a single resource beyond a certain point. Analyzing the cost-effectiveness of resource allocation is essential to achieve an optimal solution within budget constraints.

7. Conclusion

This study demonstrates the effectiveness of mathematical optimization in addressing patient waiting time challenges in emergency departments (EDs). By leveraging the MIMIC-IV-ED dataset and implementing a nurse allocation model, we successfully minimized waiting times while adhering to operational constraints such as staffing limits and area-specific requirements. The results reveal a clear relationship between the number of nurses and patient waiting times, with diminishing marginal returns as the number of nurses increases beyond 80. This emphasizes the importance of resource prioritization, particularly in time-intensive stages like observation.

Despite its simplified assumptions, such as fixed time slots and static nurse assignments, the model provides actionable insights for improving resource utilization in EDs. Future improvements, including dynamic time slot allocation, flexible nurse assignments, and incorporation of shift scheduling, could further enhance the model's practicality and impact. These findings contribute to the ongoing discourse on optimizing healthcare operations and provide a foundation for scalable solutions to improve patient care and system efficiency.

References

1. Paling, S., Lambert, J., Clouting, J., González-Esquerré, J., & Auterson, T. (n.d.). Waiting times in emergency departments: Exploring the factors associated with longer patient waits for emergency care in England using routinely collected daily data. *Emergency Medicine Journal*.
2. Robinson, J., Porter, M., Montalvo, Y., & Peden, C. J. (n.d.). Losing the wait: Improving patient cycle time in primary care. *BMJ Open Quality*.
3. Ministry of Health, Singapore. (n.d.). Attendances at emergency medicine departments. Retrieved from <https://www.moh.gov.sg/others/resources-and-statistics/healthcare-institution-statistics-attendances-at-emergency-medicine-departments/>
4. Ministry of Health, Singapore. (n.d.). Registered and enrolled nurses for nurse-to-patient ratios in general wards and in intensive care units. Retrieved from <https://www.moh.gov.sg/newsroom/registered-and-enrolled-nurses-for-nurse-to-patient-ratios-in-general-wards-and-in-intensive-care-units>